CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Pipelining Processing

Lecture No.- 3



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Recap of Previous Lecture



Topic Pipeline Throughput

Topic Latency

Topic Instruction Pipeline

Topics to be Covered



- **Topic** Pipeline Hazards
- **Topic** Data Hazard
- **Topic** Operand Forwarding



Topic : Instruction Pipeline

IF: Instruction Fetch
ID: Instruction Decode & Address Calculation
OF: Operand Fetch
EX: Execution
WB: Write Back



Topic : Instruction Pipeline

Clock Cycles

Instructions

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
I1														
I2														
I3														
I4														
I5														
I6														
I7														
I8														
I9														



Topic : Pipeline Hazards

Situations that prevent the next instruction from being executing during its designated clock cycle



Topic : Pipeline Hazards

1. Structural Hazard / Resource Conflict
2. Data Hazard / Data Dependency
3. Control Hazard / Branch Difficulty



Topic : Structural Hazard

2 different instructions try to use same resource at same time.

ex:- MUL ⇒ 2 cycles in EX phase
ADD ⇒ 1 cycle —11— ——

MUL IF ID OF EX EX WB
ADD IF ID OF - EX WB



Topic : Data Hazard or Data Dependency

Result of an instruction is used as input in next

$R1 \leftarrow R2 + R3$

$R6 \leftarrow R1 * R5$

IF ID of Ex WB

IF ID (– –) OF Ex WB

stalls

OF of dependent inst^n should be done after WB of source inst^n.

Soln of data hazard

S/ω soln (by compiler)

→ Delayed load

Compiler inserts independent or no-operation instructions before dependent inst^n to delay it's OF.

H/ω solution

→ 1. Hardware interlock

→ 2. operand forwarding (bypassing)

Delayed load :-

R1 ← R2 + R3

R6 ← R1 * R5

↓

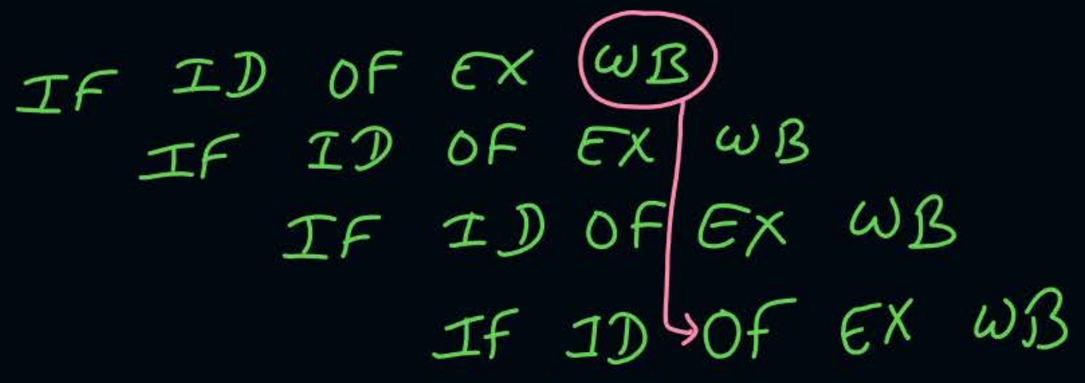
delayed load

R1 ← R2 + R3

Instⁿ 1

Instⁿ 2

R6 ← R1 * R5



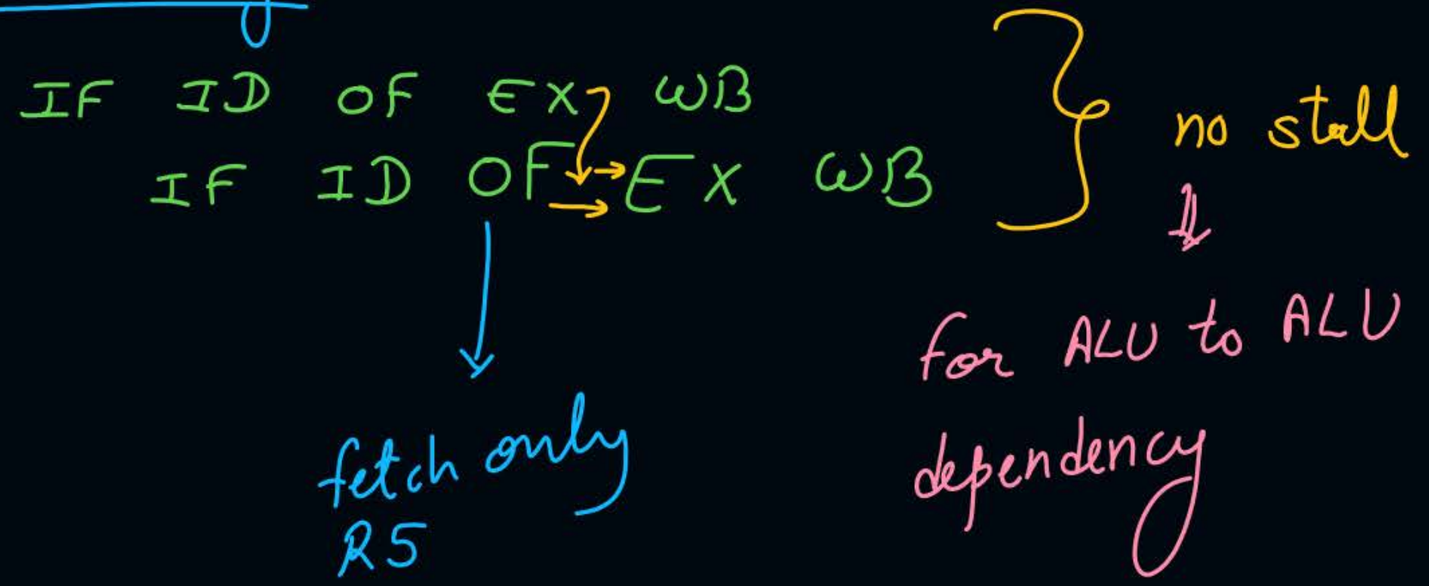
Hardware interlock :-

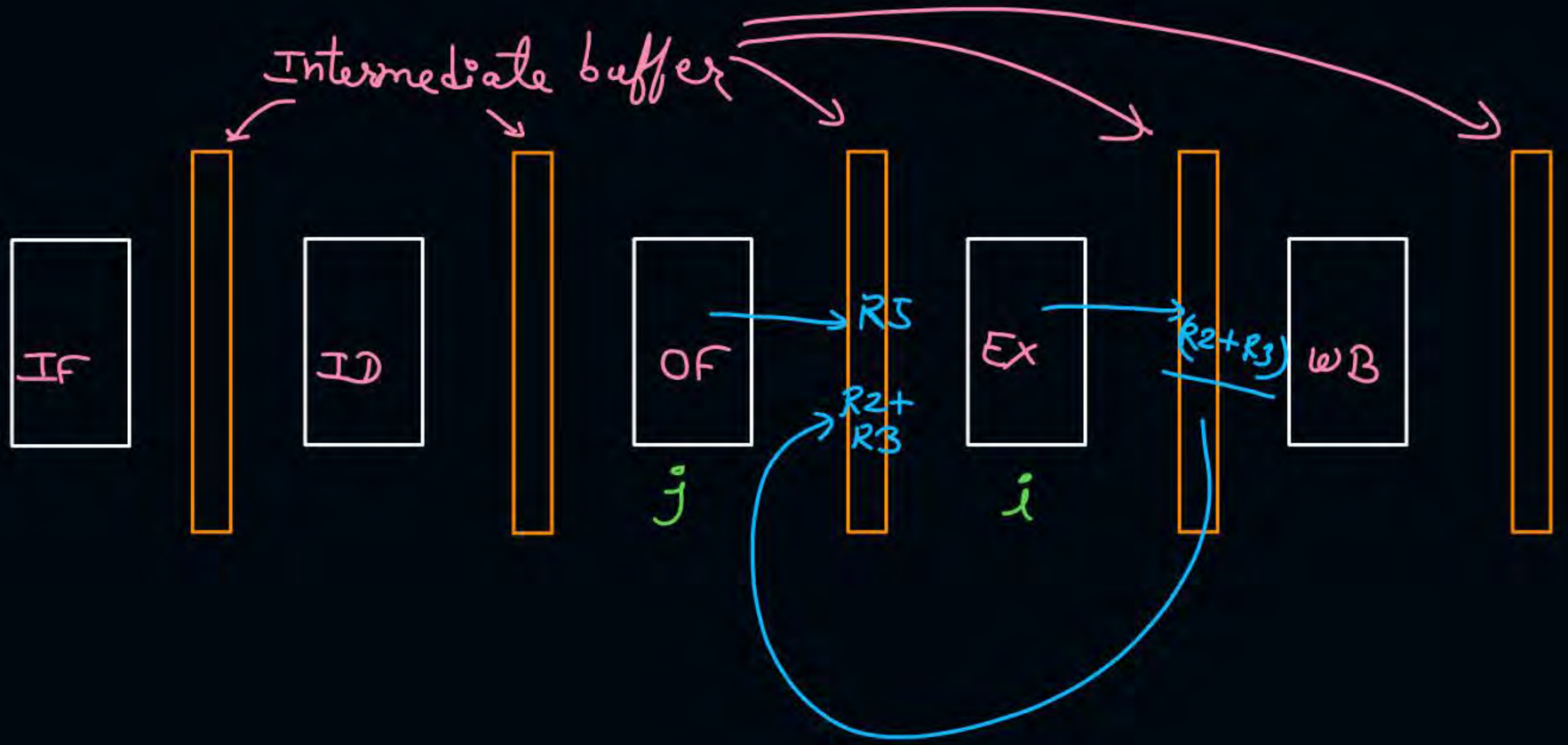
Providing stalls

Bypassing or operand forwarding :-

i R1 ← R2 + R3

j R6 ← R1 * R5




Intermediate buffer
IF
ID
OF
EX
WB
R5
R2+R3
(R2+R3)
j
i

Load inst^{n} to ALU dependency =>

$R1 \leftarrow M[1000]$

$R6 \leftarrow R1 * R5$

} stalls will be needed even if operand forwarding is used.

ALU to store dependency :-

$R1 \leftarrow R2 + R3$

$M[1000] \leftarrow R1$

}

No. of stalls due to data hazard in 2 immediate instns = WB phase no − OF phase no.

$= 5 - 3$

$= 2$

R1 ← R2 + R3
R8 ← R9 + 5
R5 ← R1 * R6

stalls = WB phase no. − OF phase no. − no. of independent instns b/w 2 dependent instns

$= 5 - 3 - 1$

$= 1$



Topic : Control Hazard or Branch Difficulty

Hazards because of branch instructions

Soln

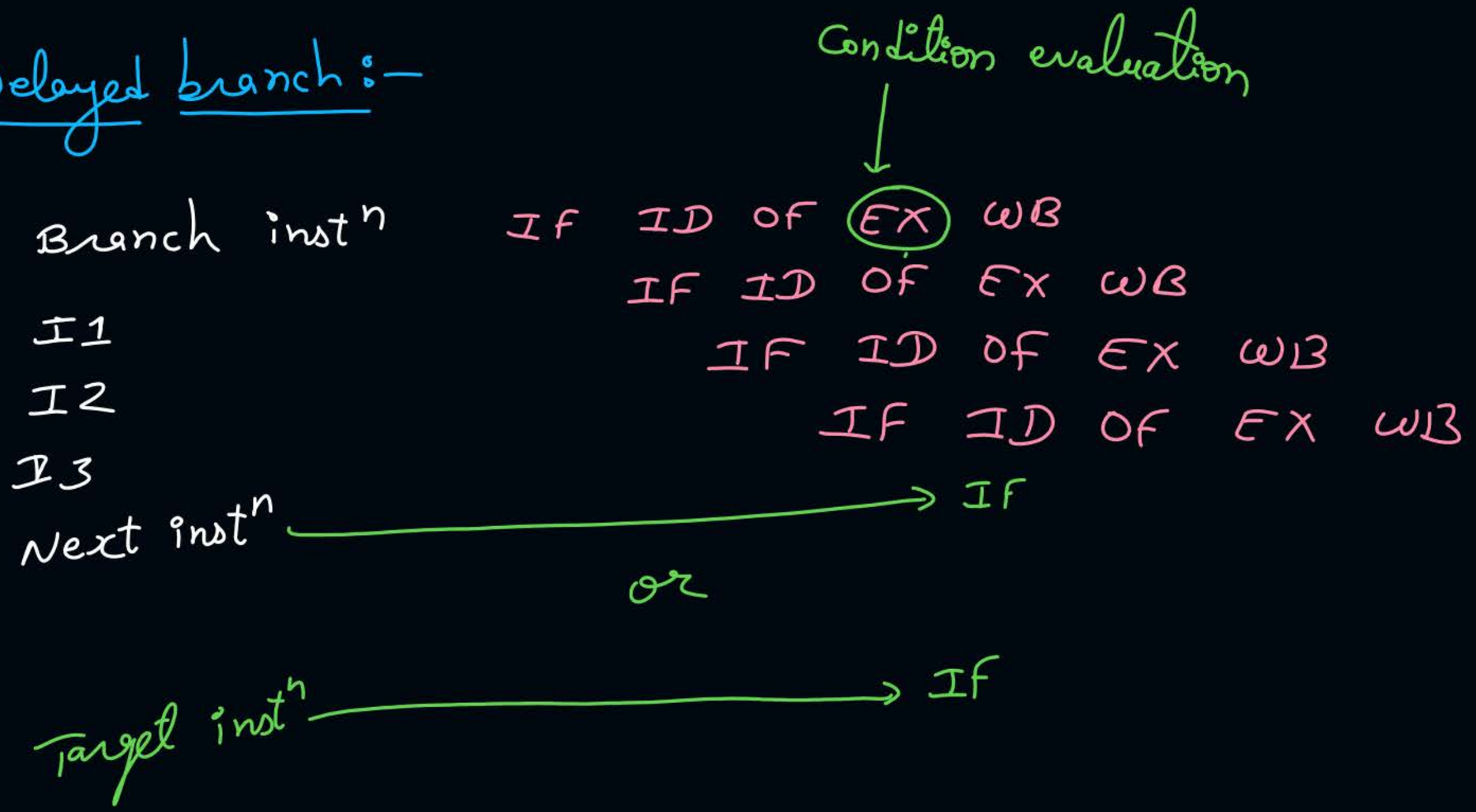
S/w soln
(by compiler)

→ Delayed branch

H/w soln

→ Branch prediction

Delayed branch :-



Branch prediction :-

↓

H/w predicts branch will be taken or not and starts executing further inst^ns accordingly.

[NAT] GATE- 2008

Ans = 8

#Q. Consider a pipelined processor with the following four stages:

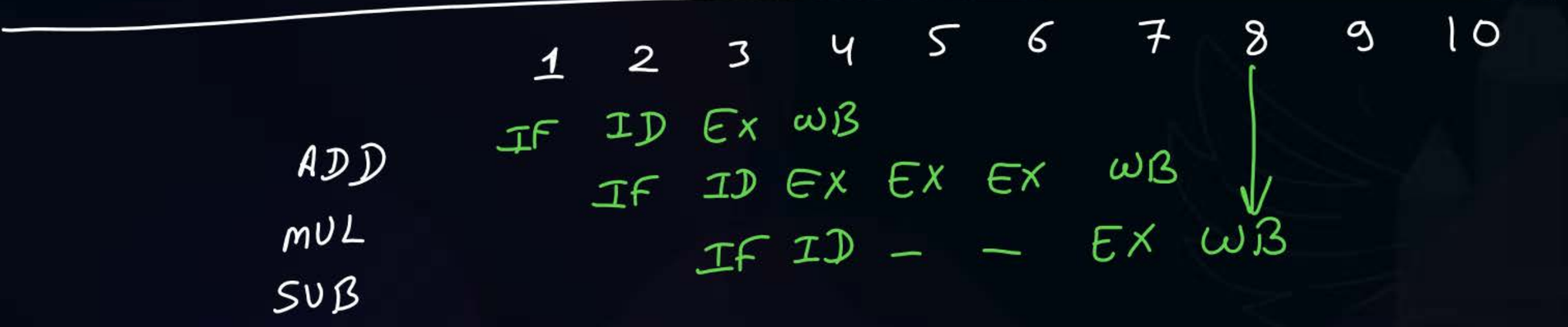
1) IF: Instruction Fetch
2) ID: Instruction Decode and Operand Fetch
3) EX: Execute
4) WB: Write Back

The IF, ID, and WB stages take one clock cycle each to complete the operation. The number of clock cycles for the EX stage depends on the instruction. The ADD and SUB instructions need 1 clock cycle and the MUL instruction needs 3 clock cycle in the EX stage. Operand forwarding is used in the pipelined processor. What is the number of clock cycles taken to complete the following sequence of instructions?

ADD R2, R1, R0	R2←R1+R0
MUL R4, R3, R2	R4←R3*R2
SUB R6, R5, R4	R6←R5-R4

[Solution]


IF
ID
EX
WB
1
1
1
ADD SUB MUL
1 1 3
extra 0 0 2


1 2 3 4 5 6 7 8 9 10
ADD
MUL
SUB
IF ID Ex WB
IF ID EX EX EX WB
IF ID — — EX WB

w/o hazard no of cycles = $k + n - 1 = 4 + 3 - 1 = 6$

stalls due to structural hazard = 2

Total = 8

Ques) In prev quest^{n} if seq.

ADD
MUL
SUB
MUL

no. of cycles ? Ans = 11

Soln

w/o stalls = $k + n - 1 = 4 + 4 - 1 = 7$

stalls due to structural hazard = $2 * 2 = 4$

Total = 11

[NAT]

Ans = 29

#Q. In previous question if there are 7 ADD, 4 SUB and 5 MUL instructions then What is the number of clock cycles taken to complete such sequence of instructions?

$n = 16$

w/o stalls = 4 + 16 − 1 = 19

stalls = 5 * 2 = 16

Total = 29

Ques) IF — 1
ID — 1
OF — 1
EX — ?
WB — 1

	ADD	SUB	MUL	DIV
EX	1	2	3	6
stalls	0	1	2	5

Instⁿ seq:- MUL
SUB
ADD
DIV

no. of cycles = 16 ?

w/o stalls = 5 + 4 − 1 = 8
stalls = 1 + 2 + 5 = 8

Total = 16 Ans.

[NAT]

stalls for data hazard also

#Q. In previous question if no operand forwarding is used what is the number of clock cycles taken to complete the given sequence of instructions?

ADD R2, R1, R0	R2←R1+R0
MUL R4, R3, R2	R4←R3*R2
SUB R6, R5, R4	R6←R5-R4

2 data dependency

stalls for each data dependency
= 4-2 = 2

w/o hazard = 4+3-1 = 6
stalls for structural hazard = 2
stalls for data hazard = 2*2 = 4
Total = 12

Ques) In prev. ques^n speed up achieved by using operand forw. as compared to not using operand forwarding

Soln

$$= \frac{12}{8}$$

$$= 1.5$$

[NAT] GATE-PYQ 2018

Ans = 219

#Q. The instruction pipeline of a RISC processor has the following stages: Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Writeback (WB), The IF, ID, OF and WB stages take 1 clock cycle each for every instruction. Consider a sequence of 100 instructions. In the PO stage, 40 instructions take 3 clock cycles each, 35 instructions take 2 clock cycles each, and the remaining 25 instructions take 1 clock cycle each. Assume that there are no data hazards and no control hazards.

The number of clock cycles required for completion of execution of the sequence of instruction is 219 ?

IF — 1
ID — 1
OF — 1
PO
WB — 1

40 inst^{n}	35 inst^{ns}	25 inst^{ns}
3	2	1
stalls = 2 * 40 = 80	= 1 * 35 = 35	0

Total stalls = 115

w/o hazard = 5 + 100 − 1 = 104

stalls = 115

219

[NAT] GATE- PYQ

Ans = 23

#Q. Consider a 4-stage pipeline processor. The number of cycles needed by the four instructions I1, I2, I3, I4 in stages S1, S2, S3, S4 is shown below:

	S_1	S_2	S_3	S_4
I1	2	1	1	1
I2	1	3	2	2
I3	2	1	1	3
I4	1	2	2	2

What is the number of cycles needed to execute the following loop?

For (i = 1 to 2) (I1; I2; I3; I4)

[Solution]

	S_1	S_2	S_3	S_4
I1	2	1	1	1
I2	1	3	2	2
I3	2	1	1	3
I4	1	2	2	2

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
I1	S1	S1	S2	S3	S4									
I2			S1	S2	S2	S2	S3	S3	S4	S4				
I3				S1	S1	–	S2	–	S3	–	S4	S4	S4	
I4														
I1														
I2														
I3														
I4														

[Solution]

	S_1	S_2	S_3	S_4
I1	2	1	1	1
I2	1	3	2	2
I3	2	1	1	3
I4	1	2	2	2

	S1	S2	S3	S4
I1	2	3	4	5
I2	3	6	8	10
I3	5	7	9	13
I4	6.	9	11	15
I1	8	10	12	16
I2	9	13	15	18
I3	11	14	16	21
I4	12	16	18	23



2 mins Summary

Topic Pipeline Hazards

Topic Data Hazard

Topic Operand Forwarding



Happy Learning

THANK - YOU